THE THREE DESCRIPTIONS OF THE COHERENT K-GROUP ON A NOETHERIAN SCHEME AGREE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The zeroth K-group of an abelian category equals that of its bounded derived category. invoked at proof lines 10092-10093. Gives the first equality.
- The K-group of a bounded derived category with coherent cohomology equals the K-group of the abelian category of coherent sheaves. invoked at proof lines 10094-10096. Gives the second equality.
- On a Noetherian scheme D^b(Coh(O_X)) equals D^b_{Coh}(O_X). recalled at proof lines 10098-10101. Alternative argument for the second equality.

Source reference. Original lemma 0FCP: sources/stacks-chow/tags.txt, source lines 19189–19189. Cited at proof lines 10092-10093 for the first equality; the second equality is inside the counted span.

Source reference. Original lemma 0FCS: sources/stacks-chow/tags.txt, source lines 19192–19192. Cited at proof lines 10095-10096 for the second equality; the alternative route is inside the counted span.

Source reference. Original proposition 0FDB: sources/stacks-perfect/pinned-original-source.tex, source lines 2813–2813. Cited at proof lines 10100-10101 as the alternative route for the second equality; the first equality is inside the counted span.

Lemma 0.2. *Let X be a Noetherian scheme. Then*

$$K_0(Coh(\mathcal{O}_X)) = K_0(D^b(Coh(\mathcal{O}_X)) = K_0(D^b_{Coh}(\mathcal{O}_X))$$

Proof. The first equality is Derived Categories, Lemma 0FCP. We have $K_0(Coh(\mathcal{O}_X)) = K_0(D^b_{Coh}(\mathcal{O}_X))$ by Derived Categories, Lemma 0FCS. This proves the lemma. (We can also use that $D^b(Coh(\mathcal{O}_X)) = D^b_{Coh}(\mathcal{O}_X)$ by Proposition 0FDB to see the second equality.) □

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